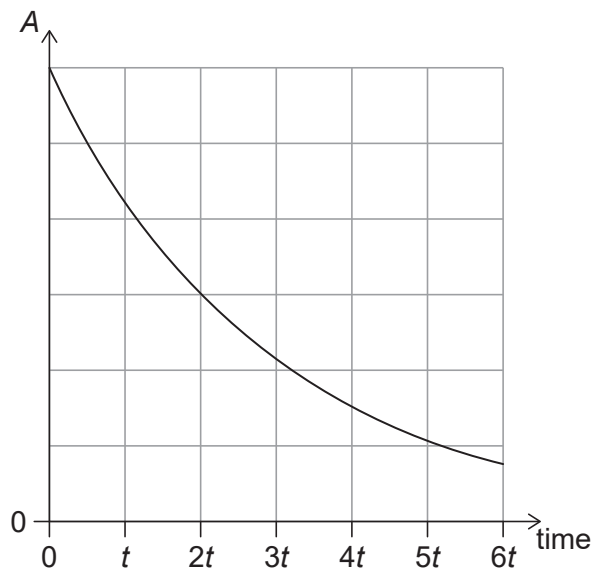


E.3 / PAPER 1 / QUESTION 24

28 April 2022 / TZ2 / HL Paper 1 / original Q25 / 2222-6513

24. A radioactive nuclide X decays into a nuclide Y. The graph shows the variation with time of the activity A of X. X and Y have the same nucleon number.



What is true about nuclide X?

- A. alpha (α) emitter with a half-life of t
- B. alpha (α) emitter with a half-life of $2t$
- C. beta-minus (β^-) emitter with a half-life of t
- D. beta-minus (β^-) emitter with a half-life of $2t$